Symmetries in Physics — Exercise Sheet 5

Characters, tables, and projectors

Variant: Questions only

Exercise 1 Construct the character table of S_3 from first principles: find the classes, write down the two one-dimensional characters, and determine the third row using orthogonality alone. Verify column orthogonality afterwards.

Exercise 2 Compute the character tables of D_4 and of Q_8 and show they are *identical* (after matching classes). Conclude that the character table does not determine the group.

Exercise 3 S_4 permutes the coordinates of $\mathbb{C}^4$. Compute the character of this permutation representation, then use the character calculus to decompose it and to prove that its three-dimensional summand is irreducible.

Exercise 4 For the cyclic action of $\mathbb{Z}/3\mathbb{Z}$ on $\mathbb{C}^3$ (Sheet 3, Exercise 1), write down the three character projectors $P_k = \frac{1}{3}\sum_g \overline{\chi_k(g)}\,\Pi(g)$ and apply each to e_1; identify the images.

Exercise 5 (retrieval: parity through character projectors) Return to the parity action of $\mathbb{Z}/2\mathbb{Z}$ on $L^2(\mathbb{R})$ from Sheet 3, Exercise 5. Using the two characters and the character-projector formula from this lecture, compute both projectors and identify their images. Explain why each image is an isotypic component but is not itself an individual irreducible representation.

Exercise 6 The *tensor square* of a representation Π on V is the action $(\Pi\otimes\Pi)(g) = \Pi(g)\otimes\Pi(g)$ on $V \otimes V$; its *symmetric* and *antisymmetric squares* are the restrictions to the invariant subspaces $\mathrm{Sym}^2 V$ and $\Lambda^2 V$, spanned by the $e_i e_j + e_j e_i$ $(i \le j)$, respectively the $e_i e_j - e_j e_i$ $(i < j)$. (So if $\Pi(g)$ has eigenvalues $\{\lambda_i\}$, then $(\Pi \otimes \Pi)(g)$ has eigenvalues $\{\lambda_i\lambda_j\}_{i,j}$.) (i) If Π has eigenvalue multiset $\{\lambda_i\}$ at g, show that these have characters

$$\chi_{\mathrm{Sym}^2}(g) = \tfrac{1}{2}\big(\chi(g)^2 + \chi(g^2)\big), \qquad \chi_{\Lambda^2}(g) = \tfrac{1}{2}\big(\chi(g)^2 - \chi(g^2)\big).$$

(ii) Apply both to the two-dimensional irreducible E of S_3 and decompose the results.

Exercise 7 Let Π be a d-dimensional representation of a finite group. Show $|\chi(g)| \le d$ for every g, with equality if and only if $\Pi(g)$ is a scalar multiple of the identity. Deduce that $\chi(g) = d$ exactly on the kernel of Π: the character detects faithfulness.